

Understanding Technical Interviews as Pedagogy for Computer Science Students

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ABSTRACT

Computer Science (CS) graduates often lack the industry-relevant competencies [4, 8, 9] that employers assess through technical interviews. Moreover, technical interviewing is a skill in its own right that, despite requiring weeks of dedicated practice to learn [1, 7], is rarely taught in CS curricula. This has resulted in inequitable preparation burdens for students [6, 7]. Motivated by students' need to develop technical interviewing skills and CS departments' desire to both assess and ensure mastery of industry-relevant competencies, my proposed dissertation work argues that technical interviews can serve dual pedagogical purposes: as a practical tool for industry preparation and as a mechanism for assessing and developing industry-relevant competencies in CS students at scale.

I propose technical interviews as pedagogy for CS students through three interconnected studies: 1) deriving industry-relevant competencies from a qualitative analysis of feedback given by professional programmers during 167 mock interviews, 2) a co-designed video-annotation tool that scaffolds students' reflection on their mock interview performance and helps them develop a study plan, and 3) a course design that incorporates peer mock interviews and structured reflection via the tool developed in Study 2, with assessments and learning outcomes informed by the empirically-derived competencies from Study 1.

With this work, I plan to contribute empirically-grounded specifications of industry-relevant competencies, pedagogical tools for scaffolding technical interview preparation and reflection, and a course design that uses mock interviews to position students as both learners and "more knowledgeable others" [10] during the technical interview process.

CCS CONCEPTS

• Social and professional topics→Professional topics~Computing education; Computing profession

KEYWORDS

academia-industry gap, computer science education, industry-relevant education, technical interviews

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1 Questions and Connections

As a dual-enrolled pre-candidate, my immediate goal for this consortium is to receive feedback on the progress I have made as part of Study 1, which will serve as the foundation for both my Computer Science and Learning Sciences qualifying exams. So far, I have used evaluation coding to analyze the written and verbal feedback given during two mock interviews, resulting in an initial set of 33 qualitative codes. As I continue to analyze more of the mock interviews and develop this inductive codebook, I ask:

1. How can I use patterns across the codes to derive competency specifications in accordance with the suggestions from the ACM 2020 Curricula [2]?
2. Given that the feedback was generated by industry practitioners and not educators, what considerations might I need to make?

I am also looking forward to connecting with researchers and faculty who are working with competency frameworks [2, 3, 5] or are more broadly dedicated to alleviating disparity in preparation for CS students as they transition from academia to industry. I am also looking forward to hearing from peers across different PhD milestones who are similarly dedicated to thinking critically about CS as a discipline in a world where LLMs are capable of generating computer code.

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